

Tarun Emmanuel Majhi

(774) 224-0556 | tarunemmanuel10@gmail.com | LinkedIn | GitHub | Portfolio

SUMMARY

Machine Learning Engineer with around 4 years of experience designing, deploying, and maintaining production ML platforms and streaming infrastructure across banking and insurance. Specialized in real-time inference pipelines, graph neural networks, MLOps, and enterprise GenAI systems using Python, PySpark, PyTorch, AWS SageMaker/EKS, and vector databases. Proven track record of delivering high-impact financial systems, including cutting annual trade fail penalties by \$3.2M at State Street and reducing manual insurance claim review times by 35%.

TECHNICAL SKILLS

Machine Learning & AI: Machine Learning Systems Design, Gradient Boosting (XGBoost, LightGBM, CatBoost), Graph Neural Networks (PyTorch Geometric), Natural Language Processing, Transformer Models (BERT, Sentence-Transformers), Retrieval-Augmented Generation (RAG), Anomaly Detection, Time-Series & Sequence Modeling, Feature Engineering, Model Explainability (SHAP), Cost-Sensitive Learning, Imbalanced Data Management (SMOTE)

MLOps & Lifecycle: Model Deployment, Streaming & Batch Inference, Model Monitoring (Evidently AI, TruLens), Experiment Tracking (MLflow), Model Optimization (ONNX Runtime, AWQ Quantization), Vector Search Infrastructure, Feature Stores (Feast, Redis Enterprise), CI/CD for ML, Data & Concept Drift Detection

Programming Languages & Frameworks: Python, PySpark, Polars, Scikit-learn, PyTorch, LangChain, LlamaIndex, NetworkX, NumPy, Pandas, FastAPI

Distributed Computing & Cloud: AWS (SageMaker, EKS, S3, Lambda, EventBridge, Redshift), Azure (Azure ML, Databricks, Event Hubs, CosmosDB), Ray (Ray Serve, Ray Cluster), Docker

Data Engineering & Streaming: Apache Kafka, Snowflake (Iceberg Tables), Apache Airflow, PostgreSQL (pgvector), SQL, Vector Databases (Pinecone, Qdrant, FAISS, Milvus)

Domain Knowledge (BFSI): P&C Insurance Claims Fraud, Credit Risk Underwriting, Anti-Money Laundering (AML) Anomaly Detection, T+1 Trade Settlement Operations, Fixed Income Risk Attribution, Financial Document Parsing (ISO 20022, Swift)

PROFESSIONAL EXPERIENCE

Machine Learning Engineer State Street, USA	Sep 2025 - Present
- Engineered a low-latency T+1 settlement fail prediction system on AWS EKS using LightGBM and ONNX Runtime to intercept high-risk trade exceptions. Streamlined intraday routing for custody operations, cutting manual investigations by 32% and mitigating \$3.2M in annual penalty fees. - Constructed sub-50ms streaming feature pipelines via Apache Kafka, PySpark, and Redis Enterprise to ingest Swift ISO 20022 message flows and custodian balance feeds. Enabled real-time liquidity exposure scoring that achieved an 89% precision rate prior to market cutoffs. - Deployed multi-task neural network models within Ray Serve to simultaneously forecast settlement failure probabilities and estimate overnight cash buffer requirements. Automated risk-prioritized ticket queues inside the State Street Alpha middle-office platform. - Architected a fixed-income similarity framework using PyTorch Geometric Graph Neural Networks to model complex corporate bond capital structures. Generated 256-dimensional embeddings for 1.2M+ instruments, improving missing price estimation accuracy by 24% RMSE. - Scaled a high-throughput Qdrant vector search infrastructure on Ray clusters to execute sub-10ms nearest-neighbor queries across illiquid municipal and corporate assets. Accelerated quantitative factor-risk calculations for SSGA portfolio managers from 45 minutes down to 4. - Formulated sparse regularized ensemble algorithms paired with Kalman Filtering to deliver dynamic factor risk attribution matrices for non-liquid securities. Enhanced portfolio stress-testing capabilities across 5,000+ institutional client accounts.	

Machine Learning Engineer Happiest Minds Technologies, India	Apr 2021 - Dec 2023
- Accelerated Scikit-Learn and TensorFlow model-training workflows by optimizing Pandas transformations and feature-matrix construction, reducing average experimentation runtime by 31% across 20+ model iterations. - Engineered an end-to-end P&C claims fraud detection model on AWS SageMaker using XGBoost and CatBoost to replace legacy rule-based intake. Handled extreme class imbalance with SMOTE-Tomek, cutting manual reviews by 35% and saving \$1.8M annually. - Integrated NLP sentiment analysis on First Notice of Loss (FNOL) adjuster notes using BERT embeddings, feeding risk signals directly into the scoring engine. Accelerated straight-through processing for clean claims while increasing early fraud capture by 18%. - Architected a real-time credit underwriting pipeline using PySpark, Azure Databricks, and Feast feature store to analyze streaming transaction velocity. Boosted default prediction accuracy by 12% over traditional scoring methods. - Implemented SHAP explainability frameworks at inference to generate dynamic regulatory reason codes for automated loan decisions. Streamlined FCRA compliance workflows and reduced false-positive AML transaction alerts by 28%. - Built a Hybrid-RAG document retrieval system using LangChain, Pinecone, and Azure OpenAI to parse complex commercial insurance policy PDFs. Reduced underwriter search times from 15 minutes to under 10 seconds per query. - Designed a cross-encoder reranking pipeline with strict system guardrails to prevent LLM hallucinations during customer support interactions. Improved agent issue resolution rates by 40% while ensuring 100% factual accuracy against policy documents.	

EDUCATION

Clark University Worcester, MA	Jan 2024 – Dec 2025
MS in Computer Science, GPA: 3.9/4.0	

PROJECTS

Stock Market Trend Prediction Analysis with Deep Learning and ML
- Engineered a high-frequency trading forecasting engine using LSTM and GRU networks with TensorFlow, achieving a 25% increase in prediction accuracy for historical market datasets. - Developed a full-stack financial dashboard using FastAPI and React.js to visualize real-time inference, reducing latency by 30% through optimized preprocessing pipelines and NumPy vectorization.
RAG Assistant for SAP Incident Troubleshooting
- Developed a RAG-based assistant that ingests historical SAP tickets, dumps, job logs, and internal KB/SAP Notes into a vector database for similarity search. - Implemented text chunking, embeddings, and retrieval pipelines so that for a new incident, the system surfaces similar cases and contextual documents, enabling LLMs to generate targeted troubleshooting guidance.